

PHASE 6 – PROJECT TESTING

6.1 Introduction

Testing is performed to verify that the APIs, database operations, authentication mechanisms, role permissions, and AI services work according to the defined requirements.

Postman or Thunder Client can be used to send requests to the backend and verify the returned responses.

6.2 Testing Objectives

The main objectives are:

- Verify API functionality.
- Validate database operations.
- Verify user authentication.
- Verify role-based authorization.
- Test blog CRUD operations.
- Test AI services.
- Verify input validation.
- Verify error handling.
- Identify and resolve defects.

6.3 API Test Cases

Test Case	Test Scenario	Expected Result	Status
TC001	User Registration	New user should be registered	Pass
TC002	User Login	Valid credentials should generate authentication response	Pass
TC003	Invalid Login	Invalid credentials should be rejected	Pass
TC004	Create Blog	Authorized user should create a blog	Pass
TC005	Fetch Blogs	Published/available blogs should be retrieved	Pass
TC006	Update Blog	Authorized user should update permitted blog data	Pass
TC007	Delete Blog	Authorized user should delete permitted content	Pass
TC008	Add Comment	Valid user should be able to submit a comment	Pass
TC009	Generate AI Blog	AI service should return generated content	Pass
TC010	Summarize Blog	AI service should return a summary	Pass
TC011	Unauthorized Request	Restricted request should be rejected	Pass

TC012	Invalid Input	Invalid request data should return an appropriate error	Pass
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6.4 Security Testing

Security testing verifies:

- JWT token validation.
- Password protection.
- Role-based access.
- Unauthorized request rejection.
- Input validation.
- Input sanitization.

6.5 Database Testing

Database testing verifies:

- Successful MongoDB connection.
- Correct document creation.
- Correct document retrieval.
- Data modification.
- Data deletion.
- Schema validation.

6.6 AI Feature Testing

AI features are tested by providing different content prompts and verifying that the backend receives and returns the generated AI response correctly.

6.7 Testing Result

The testing phase confirms the major backend operations and verifies that the application behaves according to the defined functional requirements.



